ANURAG MITRA

"Driving AI-Powered Risk Intelligence & Business Transformation through Data Science Leadership"

+91 – 9873341286

anurag.mitra54@gmail.com

Senior Data Science professional aiming to lead AI-powered business transformation through data-driven strategy, model governance, and scalable ML deployment. Committed to fostering collaboration between business and technology teams to unlock insights, manage risk, and accelerate growth.

CORE COMPETENCIES

- AI Strategy & Transformation
- Enterprise Risk Modeling & Governance
- Generative & Agentic AI Systems
- End-to-End Machine Learning Lifecycle Management
- Regulatory Compliance & Explainable AI
- Big Data Engineering & Distributed Computing
- Cloud-Native AI Infrastructure & CI/CD Automation
- AI Ethics, Fairness & Responsible AI Practices
- Data Quality, Governance & Lineage Management
- NLP & Document Intelligence Solutions

TECHNICAL SKILLS

- **Programming & Analytics:** Python | SQL | Predictive Modeling | Machine Learning | Deep Learning
- **AI & Advanced Systems:** Generative AI | Agentic AI | MLOps | DataIku
- **Cloud & Big Data:** AWS | Azure | PySpark
- **Visualization & Insights:** Tableau | Power BI

EDUCATION

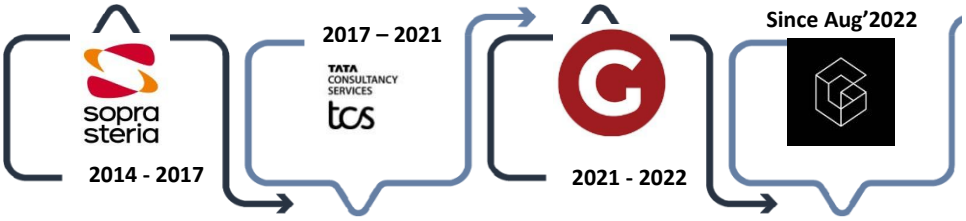
Master of Computer Applications (MCA), Pondicherry University, Puducherry |Jun'2011 – May'2014

Bachelor of Science (B.Sc) in Computer Science, St. Xavier's College, Kolkata, Jun'2007 – May'2010

PROFILE SUMMARY

- **Accomplished AI & Data Science Leader** with 11+ years of experience driving enterprise-scale Machine Learning, Risk Analytics, and Financial Crime Prevention across global banking and fintech ecosystems.
- **Strategic innovator in architecting and operationalizing** AI-driven solutions, leveraging Predictive, Generative, and Agentic AI to deliver measurable business impact and regulatory compliance.
- **Proven expertise in Fraud Detection, AML Modeling,** and Transaction Surveillance, with deep proficiency in identifying, quantifying, and mitigating emerging financial crime risks.
- **Hands-on leader in designing and deploying end-to-end ML systems**, integrating advanced algorithms, scalable data pipelines, and automated MLOps workflows across AWS and Azure environments.
- **Visionary AI strategist skilled in developing data-driven roadmaps,** model governance frameworks, and explainability standards aligned with business, risk, and regulatory mandates.
- **Expert in advanced modeling techniques — Logistic Regression,** XGBoost, SVM, DNNs, RNNs — for risk scoring, behavioral analytics, and portfolio optimization.
- **Established excellence in AI democratization and Center of Excellence (CoE) leadership,** mentoring data science teams to translate analytical insights into strategic decision-making.
- **Spearheaded the deployment of real-time monitoring, model drift detection,** and automated alert triage systems, enhancing operational efficiency and reducing false positives.

CAREER TIMELINE



WORK EXPERIENCE

Genpact India Pvt. Ltd., Bengaluru - Data Science Senior Manager | Aug'2022 — Present

Key Result Areas:

- Spearheaded the design and deployment of AML-focused Machine Learning models across global banking portfolios, enhancing the precision of high-risk jurisdiction monitoring and improving compliance effectiveness.
- Engineered an Excessive Activity Detection System leveraging ML-based behavioral analytics to flag anomalous cash transactions, driving early risk identification and reducing false positives by ~30%.
- Redesigned and productionized the Structuring Model, incorporating multi-channel transactional signals (cash, EMT, monetary instruments) to strengthen layered transaction detection and improve model precision.
- Led the enhancement of the Canadian Wire Surveillance Model, applying advanced anomaly detection techniques to uncover high-value & complex wire transfer patterns indicative of potential money laundering.
- Developed an AI-driven Alert Triage & Analytics Framework integrating explainable ML and rank-scoring logic, empowering Financial Investigation Units (FIUs) to prioritize alerts and accelerate case resolution.
- Championed end-to-end model lifecycle governance, from data sourcing and feature engineering to validation, deployment, and continuous monitoring, ensuring auditability and regulatory compliance.
- Collaborated cross-functionally with risk, compliance, and data engineering teams to operationalize ML insights at scale, transforming manual investigative processes into AI-augmented decision systems.
- Responsible & Explainable AI Governance: Championed implementation of responsible AI practices across risk analytics models, embedding explainability frameworks (LIME, SHAP, and Counterfactual Analysis) to ensure transparency, compliance with global regulatory standards, and stakeholder trust in automated decision systems.
- AI-Driven Process Optimization & Cost Reduction: Spearheaded AI-led automation initiatives using Generative and Predictive AI, achieving 25–30% reduction in manual review time across financial crime investigations by integrating intelligent triage systems and self-learning feedback loops.

CERTIFICATIONS

- Applied Natural Language Processing, CMI
- Practical Machine Learning with TensorFlow, IIT Madras – Google
- Introduction to Machine Learning, IIT Kharagpur

PERSONAL DETAILS

Date of Birth: 29th August 1989

Languages Known: English, Hindi, Bengali

Address: Bangalore, India

- Enterprise Data Strategy & Model Lifecycle Management: Architected scalable data governance and model lifecycle frameworks integrating feature stores, drift monitoring, and automated retraining pipelines — ensuring end-to-end traceability, reproducibility, & sustained model performance in production.
- Cross-Functional AI Evangelism & Mentorship: Led enterprise-wide knowledge sessions and hands-on AI enablement programs for 100+ professionals, fostering a culture of experimentation and upskilling across domains such as GenAI, MLOps, and LLM-driven automation.

PREVIOUS EXPERIENCE

GSPANN Technologies Inc., Pune - Lead Data Scientist | Jul'2021 — Jul'2022

Key Result Areas:

- Engineered an NLP-based Address Standardization Engine using entity extraction and rule-based normalization, achieving high data accuracy and improved match rates across disparate systems.
- Delivered Customer Complaint Analytics dashboards powered by topic modeling and sentiment analysis, enabling CX leaders to identify systemic issues and enhance satisfaction metrics.
- Built a Computer Vision Drift Monitoring Framework for quality inspection models, proactively detecting data or performance drifts to sustain model reliability over time.
- Developed a SHAP-enabled Attrition Prediction Model that provided interpretable insights into employee turnover drivers, supporting strategic HR retention initiatives.
- Designed and deployed Supply Chain Forecasting Models leveraging time-series ML to predict demand fluctuations and mitigate material shortages, improving procurement efficiency.

Tata Consultancy Services, Pune - Senior Business Analyst | Dec'2017 — Jul'2021

Key Result Areas:

- Designed and deployed ML-based Fraud Detection Systems for debit and credit card transactions, significantly improving early fraud identification and loss prevention.
- Engineered advanced feature engineering pipelines to derive behavioral and transactional features, enhancing model precision and long-term stability.
- Implemented NLP-driven Complaint Classification Models, automating text categorization and accelerating response times to customer grievances.
- Developed Deep Learning Autoencoder Architectures to uncover hidden fraud typologies across multi-channel transaction data, improving anomaly detection capability.
- Created executive-ready dashboards and visual analytics translating ML insights into business intelligence, supporting data-driven decision-making for risk and compliance teams.

Sopra Steria, Noida - Sr. Software Engineer - May'2014 — Dec'2017

Key Result Areas:

- Conducted customer segmentation using advanced clustering algorithms, identifying high-value segments to inform targeted marketing and engagement strategies.
- Performed spending behavior analytics using regression and exploratory modeling to surface key determinants of customer expenditure patterns.
- Partnered with business stakeholders to translate analytical insights into personalized campaigns, driving higher conversion and retention rates.
- Automated data processing and reporting workflows, enhancing analytical turnaround and accuracy in marketing intelligence projects.
- Supported cross-functional data science initiatives, integrating predictive analytics into strategic decision-making and customer lifecycle management.

PROJECTS EXECUTED

- **ESG Disclosure Analyzer | LangChain · RAG · Streamlit · Hugging Face**
  Built an AI app to evaluate corporate ESG disclosures against **GRI, SASB & BRSR** standards using **RAG pipelines with FAISS embeddings and cross-encoder re-ranking**; deployed on Hugging Face with an interactive KPI and citation-based insights dashboard.

- **Valve Document Intelligence | Fine-Tuned LLM · RAG · LangChain**
  Developed a **domain-adapted LLM** via two-stage fine-tuning and RAG architecture to enhance contextual comprehension and **document-level Q&A accuracy** across internal knowledge bases.

- **AI Email Assistant | Generative AI · LangChain · Gmail API · Streamlit · FastAPI**
  Designed an **end-to-end email triage assistant** with OAuth authentication, priority classification, thread summarization, and auto-drafting using **OpenAI/Anthropic APIs**; deployed on GCP with a secure Streamlit interface.

- **Code Transpiler | OpenAI API · Streamlit**
  Created an **AI-powered code translation platform** enabling real-time bidirectional conversion between **Python, C++**, and **Java**, with syntax-aware validation, execution benchmarking, and an interactive UI.